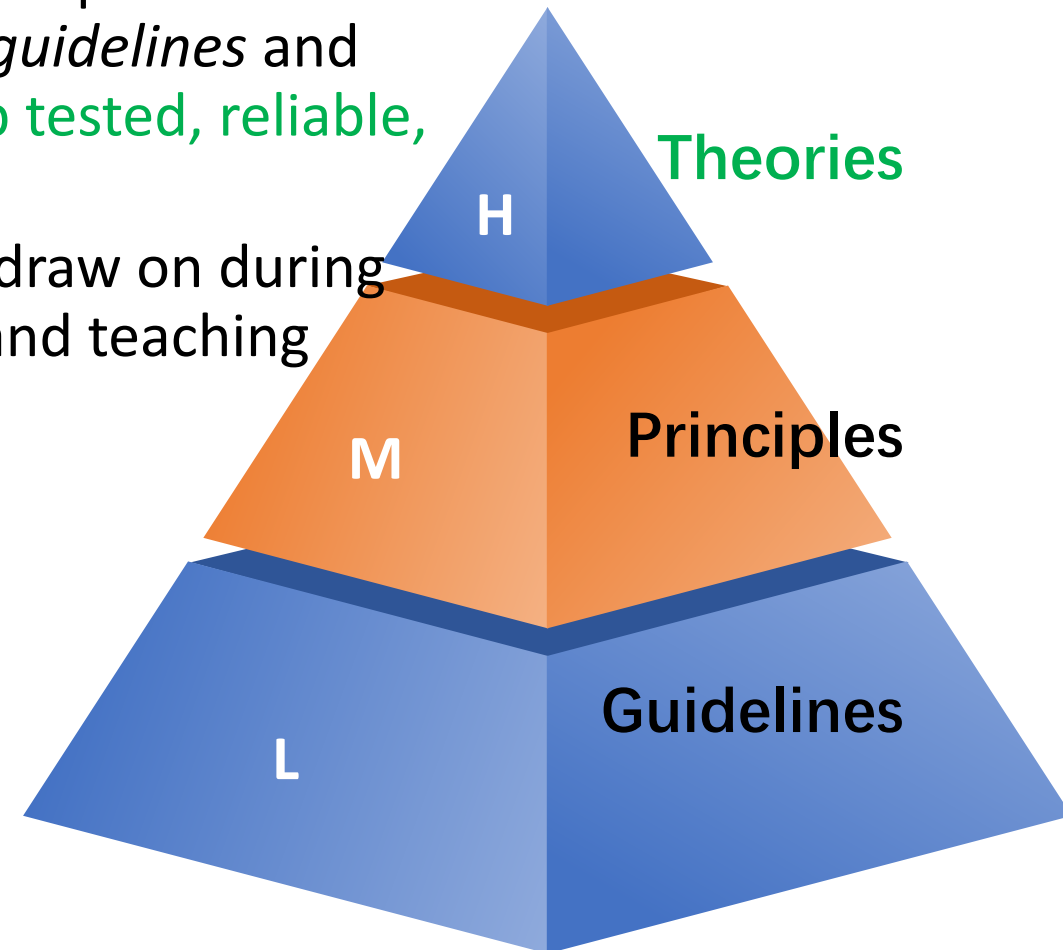


Theories

- Basics
 - One goal for the discipline of HCI (Human-Computer Interaction) is to *go beyond the specifics of guidelines* and *build on the breadth of principles* to **develop tested, reliable, and broadly useful theories**
 - High-level widely applicable frameworks to draw on during UID/evaluation to support communication and teaching
- By theory type
 - Descriptive
 - Explanatory
 - Prescriptive
 - Predictive



Theories

- Basics

- Descriptive theories

- Some theories are descriptive and these are helpful in developing consistent terminology and useful taxonomies, for objects and actions, thereby supporting collaboration/training

- Explanatory theories

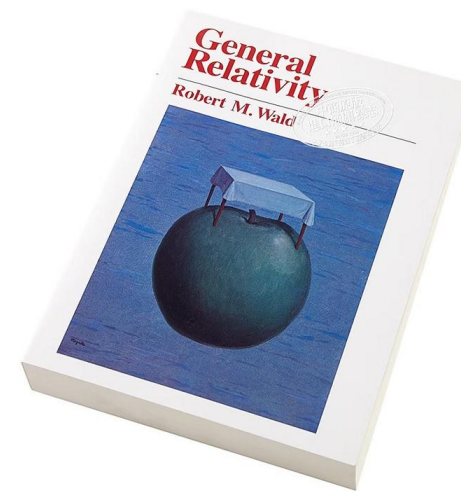
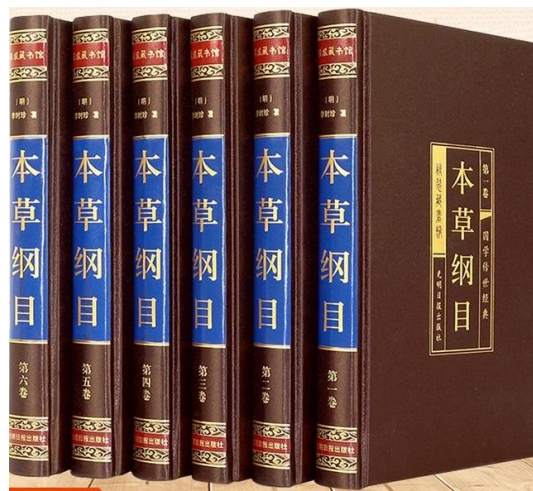
- Other theories are explanatory, describing sequences of events and, where possible, cause and effect, making interventions possible

- Prescriptive theories

- Still other theories are prescriptive, giving UIDers clear guidance for their choices

- Predictive theories

- The most precise theories are predictive, enabling UIDers to compare UI alternatives for execution time, error rates, etc.



Theories

- Basics

- High-level widely applicable frameworks to draw on during UID/evaluation to support communication and teaching

- By theory type

- Descriptive
- Explanatory
- Prescriptive
- Predictive

- By capacity/skills involved

- Motor
- Perceptual
- Cognitive

By Theory Type

- **Descriptive** Describes user interfaces and their uses with consistent terminology and taxonomies
- **Explanatory** Describes sequences of events with causal relationships
- **Prescriptive** Offers guidelines for designers to make decisions
- **Predictive** Enables comparison of design alternatives based on numeric predictions of speed or errors.

By Human Capacity

- **Motor** Skill in pointing, clicking, dragging, or other movements
- **Perceptual** Visual, auditory, tactile, and other human sensory inputs
- **Cognitive** Problem solving with short- and long-term memory

Theories

- Basics
 - High-level widely applicable frameworks to draw on during UID/evaluation to support communication and teaching
 - By theory type
 - Descriptive, Explanatory, Prescriptive, Predictive
 - By capacity/skills involved
 - Motor, Perceptual, Cognitive
 - Other *ad hoc* theories
 - Information-architecture theories
 - Subjective satisfaction prediction theories
 - Emotional reactions prediction theories
 - Small-group behaviour theories
 - Organizational dynamics theories
 -

Good theories should be understandable, produce similar conclusions for all who use it, and help to solve UID problems

Theories

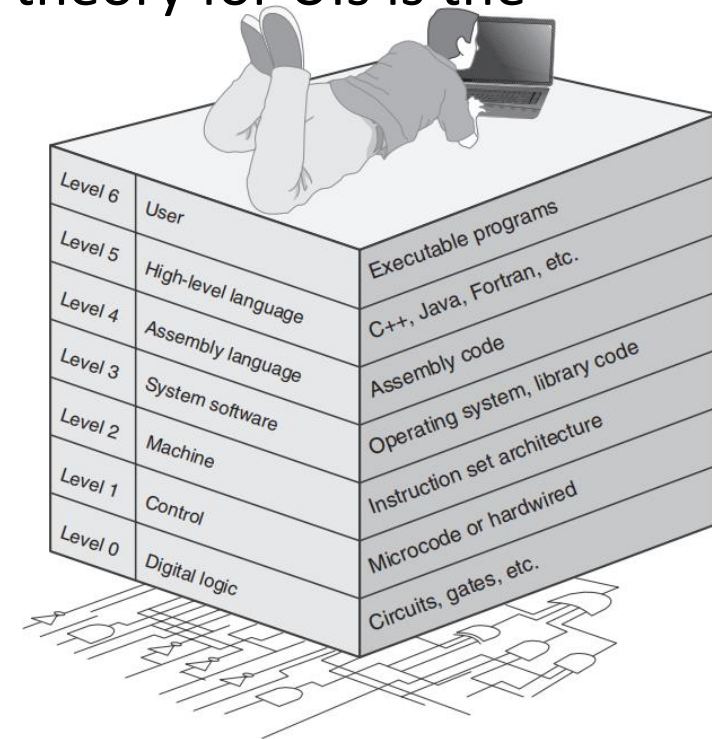
- Design-By-Levels Theories

- One approach to developing descriptive theories is to separate concepts according to levels

- Such theories have been helpful in software engineering and network design

- An appealing and easily comprehensible design-by-levels theory for UIs is the four-level theory

- Conceptual
 - Semantic
 - Syntactic
 - Lexical



Theories

- Four-Level Conceptual-Semantic-Syntactic-Lexical Theory
 - Conceptual
 - The conceptual level is the user's "mental model" of the interactive system
 - Paint programs (i.e. the first mental model for image creation) Concrete
 - They manipulate pixels and users of paint programs think in terms of sequences of actions on pixels and groups of pixels
 - Drawing programs (i.e. the second mental model for image creation) Abstract
 - They operate on objects and users of drawing programs think in terms of sequences of actions on objects and groups of objects
 - Decisions about mental models affect each of the lower levels
 - Semantic
 - Syntactic
 - Lexical

Theories

- Four-Level Conceptual-Semantic-Syntactic-Lexical Theory
 - Conceptual
 - Semantic
 - The semantic level describes the meanings conveyed by the user's input and by the computer's output display
 - Deleting an object in a drawing program could be accomplished by undoing a recent action or by invoking a delete-object action
 - Mean what is intended to mean --- Neither ambiguity nor confusion
 - Either action should eliminate only a single object and leave the rest untouched
 - Syntactic
 - Lexical

How to program a delete-object action in terms of UI primary development (instead of UI secondary development)?

Theories

- Four-Level Conceptual-Semantic-Syntactic-Lexical Theory
 - Conceptual
 - Semantic
 - Syntactic
 - The syntactic level defines how the user actions that convey semantics are assembled into complete sentences to perform certain tasks
 - The delete-files action could be invoked by dragging an object to a trash can (not 能 but 罐/桶) followed by a click in a confirmation dialog box
 - Lexical
 - The lexical level deals with device dependencies and with the precise mechanisms by which users specify the syntax
 - A function key
 - A mouse double-click within 200 milliseconds

Theories

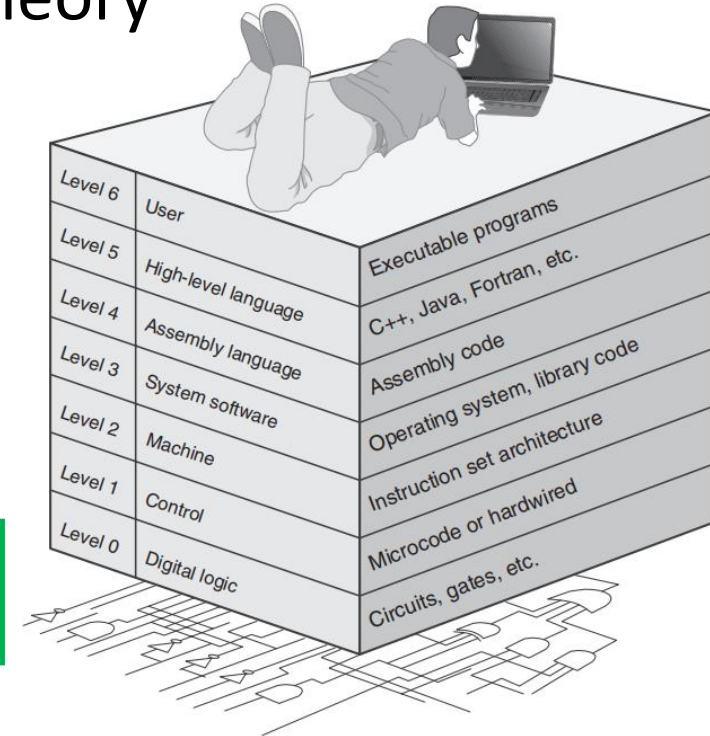
Using design-by-levels theories forces clear definitions of the high-level objects and actions, which are gathered from listening to the language used in the task domain

- Four-Level Conceptual-Semantic-Syntactic-Lexical Theory

- Conceptual
- Semantic
- Syntactic
- Lexical

- The four-level theory is convenient for UIDers because

- Its top-down nature is **easy** and **intuitive** to explain
 - Success of **graphical direct-manipulation** UIs
 - Increasingly, actions are shown by **novel visual representations**
- Its top-down nature matches the software architecture
- Its top-down nature allows for useful modularity during UID
- Successful even in more complex systems with many objects/actions
 - Most real-world objects have similar decompositions
 - Complex actions can be decomposed into smaller actions and further into atomic actions



Theories

- Stages-Of-Action Theories
 - Another approach to forming descriptive and **explanatory** theories is to portray the stages of action that users go through in using interactive products
 - Seven stages-of-action theory
 - Forming the goal
 - Forming the intention
 - Specifying the action
 - Executing the action
 - Perceiving the system state
 - Interpreting the system state
 - Evaluating the outcome

Theories

- Seven Stages-of-Action Theory

- Forming the goal
- Forming the intention
- Specifying the action
- Executing the action

- Perceiving the system state
- Interpreting the system state
- Evaluating the outcome

- Four Level Theory

- Conceptual
- Semantic
- Syntactic
- Lexical



Dynamic process of cyclic action/evaluation

Spirit of Feedback Control

Theories

- Seven Stages-of-Action Theory

- Forming the goal and intention => Specifying and executing the action



Studying errors



- Evaluating the outcome <= Perceiving and interpreting the system state

- Four principles of good UID

- The state and the action alternatives should be visible
 - There should be a good conceptual model with a consistent system image
 - The UI should include good mappings that reveal relationships among stages
 - Users should receive continuous feedback

Theories

- Consistency Theories

- An important goal for UIDers is a consistent UI
- Argument for consistency
 - If terminology for objects and actions is orderly and describable by a few rules, users will be able to learn and retain them easily

Consistent	Inconsistent A	Inconsistent B
delete/insert table	delete/insert table	delete/insert table
delete/insert column	remove/add column	remove/insert column
delete/insert row	destroy/create row	delete/insert row
delete/insert border	erase/draw border	delete/insert border

- UIDers can enforce consistency by developing detailed guidelines documents for their UIs that spell out all the consistency requirements

Theories

- Contextual Theories
- Dynamic Theories

Micro-HCI Theories	Focus on measurable performance (such as speed and errors) on multiple standard tasks taking seconds or minutes in laboratory environments
• Design-by-levels	Start with high-level design and move to smaller objects and actions
• Stages-of-action	Consider user behavior as they form intentions and seek to realize their goals.
• Consistency	Strive for consistency in objects and actions, shown by words, icons, colors, shapes, gestures, menu choices
Macro-HCI Theories	Focus on case studies of user experience over weeks and months in realistic usage contexts with rich social engagement
• Contextual	Support users who are embedded in emotional, physical, and social environments
• Dynamic	Design for evolution of user behavior as users move through levels of mastery, performance, and leadership

Theories

- Contextual Theories

- Contextual theories address the shift from use of a computer to interaction with a device-rich environment filled with sensors, responsive appliances, display walls, and audio generators
- Contextual theories often emphasize the social environment in which users are engaged with other people who can provide assistance or can be distractions
- UIDers can apply contextual theories by observing users in their own environments as they carry out their work, engage socially, or participate in sports or play

- Dynamic Theories

- Dynamic theories address UID for evolutionary development of skills mastery, behaviour change, reputation growth, and leadership capacities